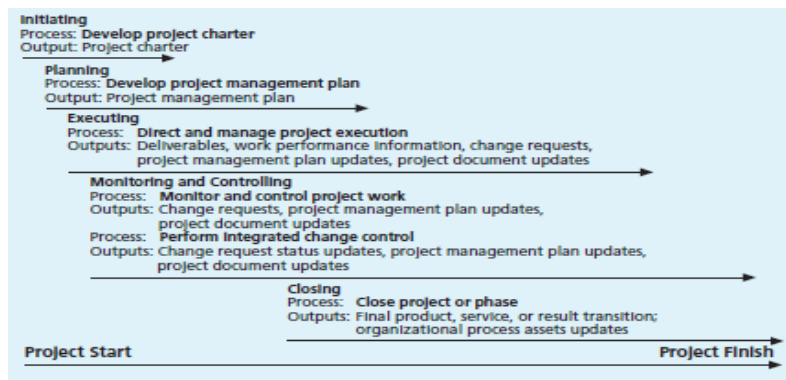


Project Initiation & Integration

Topic # 2 & 3 A
Chapter 4 – Schwalbe

Project Integration Management

- **Project integration management** involves coordinating all of the other project management knowledge areas throughout a project's life cycle.
- Project Integration Management Summary:



Project Selection

- Managers need some ways to decide which project(s) to select.
- Some ways:
 - Technical Assessment
 - Cost Benefit Analysis
 - Cash Flow Forecasting
 - Strategic Assessment

Methods for Selecting Projects

- Focusing on broad organizational needs
- Categorizing IT projects (Problems, Opportunities and directives)
- Performing net present value or other financial analysis
- Using a weighted scoring model
- Implementing a balanced scorecard

Strategic Assessment

- Not all projects can be assessed on some numeric values like quantified costs or benefits
- Assessment is based on fulfillment of strategic objectives when combined with other projects, can also take place.
- **Programmes** are formed that are collection of projects contributing to same overall organization goals.

Programmes may be.

- *Strategic*
- *Business cycle programmes*
- *Infrastructure programmes*
- *Research and development programmes*
- *Innovative partnerships*

Technical Assessment

- Evaluation whether the required functionality can be achieved with current affordable technologies.
- The cost of the technology adopted must be taken into account in the cost benefit analysis.

Limitations:

- Nature of solution produced by strategic information system plan.
- Cost of solution . Hence undergo cost benefit analysis.

Economic Assessment – Why?

- ❑ Consider whether the project is the best among other options
- ❑ Prioritise the projects so that the resources can be allocated effectively if several projects are underway

Note: A common way is to compare the expected costs of development and operation of the system with the benefits of having it in production

Economic Assessment – How

- ❑ Cost-benefit analysis
- ❑ Cash flow forecasting
- ❑ Various cost-benefit evaluation techniques
 - ❑ Net Profit, Payback Period, ROI ,NPV and IRR

Cost-benefit analysis is to compare the estimated costs of development and operation of a system with the estimated benefits of putting the system in place.

Why need **Cash flow forecasting**? It is because the excess of benefits over costs is not sufficient to justify the implementation of a proposed project.

Cost Benefit Analysis

- Cost includes:
 - Development costs (salaries and employment costs, H/w & s/w platform costs,
 - Setup costs (training, new hardware, Database conversion)
 - Operational costs (support, maintenance, hosting, license, backup)
- Benefits:
 - Direct Benefits
 - Assessable indirect benefits (increased accuracy as easy to use software)
 - Intangible benefits

Financial Considerations for Selecting Projects

- Three primary methods for determining the projected financial value of projects include
 - Net Present Value (NPV),
 - Return on investment (ROI) and
 - Payback period.

Payback Analysis

- **Payback analysis** – a technique for determining if and when an investment will pay for itself.
- **Payback period** – the period of time that will lapse before accrued benefits overtake accrued and continuing costs.

Return-on-Investment Analysis (ROI)

Return-on-Investment (ROA) analysis – a technique that compares the lifetime profitability of alternative solutions.

The ROI for a solution or project is a percentage rate that measures the relationship between the amount the business gets back from an investment and the amount invested.

Lifetime ROI = (estimated lifetime benefits – estimated lifetime costs) / estimated lifetime costs

Net profit

Year	Cash-flow
0	-100,000
1	10,000
2	10,000
3	10,000
4	20,000
5	100,000
Net profit	50,000

‘Year 0’ represents all the costs before system is operation

‘Cash-flow’ is value of income less outgoing

Net profit value of all the cash-flows for the lifetime of the application

Pay back period

This is the time it takes to start generating a surplus of income over outgoings. What would it be below?

Year	Cash-flow	Accumulated
0	-100,000	-100,000
1	10,000	-90,000
2	10,000	-80,000
3	10,000	-70,000
4	20,000	-50,000
5	100,000	50,000

Payback Period = 4.5 years

Return on investment (ROI)

$$\text{ROI} = \frac{\text{Average annual profit}}{\text{Total investment}} \times 100$$

In the previous example

- average annual profit
= 50,000/5
= 10,000
- ROI = 10,000/100,000 X 100
= 10%

Net Present Value (NPV)

- **NPV** is a method of calculating the expected net monetary gain or loss from a project by discounting all expected future cash inflows and outflows to the present point in time.
- The time value of money recognizes that a dollar today is worth more than a dollar one year from now.

Net present value

Would you rather I gave you £100 today or in 12 months time?

If I gave you £100 now you *could* put it in savings account and get interest on it.

If the interest rate was 10% how much would I have to invest now to get £100 in a year's time?

This figure is the *net present value* of £100 in one year's time

Present Value Formula

Present value – the current value of a dollar at any time in the future.

$$PV = \frac{\text{value in year } n}{(1+r)^n}$$

Where n is the number of years and i is the discount rate.

Discount rate – a percentage similar to interest rates that you earn on your savings.

In most cases the discount rate for a business is the **opportunity cost** of being able to invest money in other projects or investments

- $(1+r)^n$ In case of 10% discount rate and one year
-Discount factor = $1/(1+0.10) = 0.9091$
- In case of 10% discount rate and two years
-Discount factor = $1/(1.10 * 1.10) = 0.8294$

NPV & Payback Analysis

	A	B	C	D	E	F	G	H	I
4	Cash flow description	Year 0	Year 1	Year 2	Year 3	Year 4	Year 5	Year 6	
5	Development cost:	(\$418,040)							
6	Operation & maintenance cost:		(\$15,045)	(\$16,000)	(\$17,000)	(\$18,000)	(\$19,000)	(\$20,000)	
7	Discount factors for 12%:	1.000	0.893	0.797	0.712	0.636	0.567	0.507	
8	Time-adjusted costs (adjusted to present value):	(\$418,040)	(\$13,435)	(\$12,752)	(\$12,104)	(\$11,448)	(\$10,773)	(\$10,140)	
9	Cumulative time-adjusted costs over lifetime:	(\$418,040)	(\$431,475)	(\$444,227)	(\$456,331)	(\$467,779)	(\$478,552)	(\$488,692)	
10									
11	Benefits derived from operation of new system:	\$0	\$150,000	\$170,000	\$190,000	\$210,000	\$230,000	\$250,000	
12	Discount factors for 12%:	1.000	\$0.893	\$0.797	\$0.712	\$0.636	\$0.567	\$0.507	
13	Time-adjusted benefits (current of present value):	\$0	\$133,950	\$135,490	\$135,280	\$133,560	\$130,410	\$126,750	
14	Cumulative time-adjusted benefits over lifetime:	\$0	\$133,950	\$269,440	\$404,720	\$538,280	\$668,690	\$795,440	
15		0	1	2	3	4	5	6	
16	Cumulative lifetime time-adjusted costs + benefits:	(\$418,040)	(\$297,525)	(\$174,787)	(\$51,611)	\$70,501	\$190,138	\$306,748	
17									
18									
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Payback Analysis

Payback Period & ROI

- The Payback period for the project:
 - Payback time = $3 + (51,611/133,560)$
 - Payback time = 3.4 years.
- ROI
 - ROI = 62.8%
 - ROI per year = 10.46% per year

Internal rate of return

- Internal rate of return (IRR) is the discount rate that would produce an NPV of 0 for the project
- Can be used to compare different investment opportunities (bank, rebort system, gold, mutual fund etc)
- There is a Microsoft Excel function which can be used to calculate

Cost Benefit Evaluation Techniques

Technique	Advantages	Disadvantages
Net Profit	Simple to Use	-Does not show profit relative to size of investment -Ignores the Timing of cash flow
Payback Period	-Simple to calculate, not particular sensitive to small forecasting errors. -Give some idea of cashflow impact	-Ignores any income (or expenditure) after the payback period. -Ignores overall profitability of project
Return on Investment (ROI)	simple and easy to calculate, quite popular	-Ignores the timing of the cash flow. -Potentially very misleading because rate of return bears no relationship with the current interest rates

Cost Benefit Evaluation Techniques - NPV

Year	Cashflow Project 1	Cashflow Project 2	Discount Factor (10%) $PV = 1/(1+r)^n$	Discounted Cashflow Project 1 (cash flow * discount factor)	Discounted Cashflow Project 2
0	-100,000	-100,000	1	-100,000	-100,000
1	10,000	30,000	0.9091	9091	27273
2	10,000	30,000	0.8264	8264	24792
3	10,000	30,000	0.7513	7513	22539
4	20,000	30,000	0.683	13,660	20490
5	100,000	30,000	0.6209	62,090	18627
Net profit	50,000	50,000			
NPV				618	13,721

Though Project 1 and 2 have same initial investment and net profits but significant different in NPV value. This difference in NPV reflects, we must wait longer for the bulk of income.

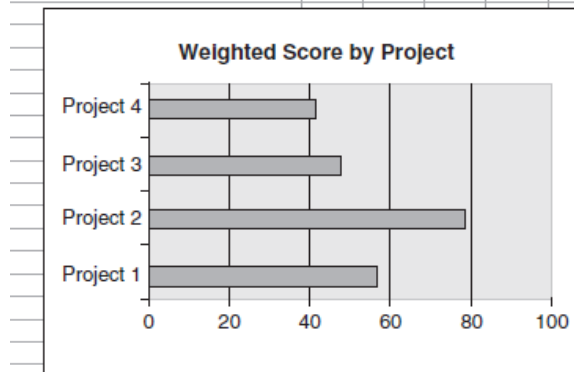
Technique	Advantages	Disadvantages
Net Present Value	-Takes into account the profitability. -Consider the timings of payments. -Consider economic situation through discount rate.	-Hard to select an appropriate discount rate. -May not be directly comparable with earnings from other investments or the costs of borrowing capital

Weighted Score Model – Selecting Projects

- A **weighted scoring model** is a tool that provides a systematic process for selecting projects based on many criteria.
- These criteria can include factors such as meeting broad organizational needs; addressing problems, opportunities, or directives; the amount of time it will take to complete the project; the overall priority of the project; and projected financial performance Of the project.

Example- Weighted Score Model

Criteria	Weight	Project 1	Project 2	Project 3	Project 4
Supports key business objectives	25%	90	90	50	20
Has strong Internal sponsor	15%	70	90	50	20
Has strong customer support	15%	50	90	50	20
Uses realistic level of technology	10%	25	90	50	70
Can be implemented in one year or less	5%	20	20	50	90
Provides positive NPV	20%	50	70	50	50
Has low risk in meeting scope, time, and cost goals	10%	20	50	50	90
Weighted Project Scores	100%	56	78.5	50	41.5



Business Drivers

- Business Driver: The root cause(s) for the project development.
- Business drivers are the crucial forces behind the successful development of a project.
- As the result one can define the [business case](#) for sustainable project development.

Critical Success Factors

- Critical success factors (CSFs), also known as Key Results Areas (KRAs), refer to the activities that must be completed to a high standard of quality in order to achieve the goals of your project.
- CSFs are a way to prioritize certain tasks as the project plan is being executed.
- Having clear CSFs helps the project team clarify what needs to be worked on first or needs special attention, allowing people to work together to achieve the project's main objectives.

Business Case

- The objective of business case is to provide a justification for the project by showing that the benefits of the project will exceed the costs of development, implementation and operation.

Feasibility Study or Project Justification – Business Case

- Introduction & Background to the proposal
- The market
- Organization & Operational Infrastructure
- Benefits
- Outline implementation Plan
- Costs
- The financial case
- Risks

Project Portfolio Management

- Project portfolio provides an overview of all the projects that an organization is undertaking or considering.
- It prioritizes the allocation of resources to projects and decides which new projects should be accepted and which should be dropped.

END OF TOPIC 2&3A

-COMING UP!!!!!!
-Project Scope Management
-WBS
-Project Team